AI-Assisted Voice Intake and Triage System

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***Abstract*—**This work illustrates the feasibility of lightweight AI-assisted triage within small-to-mid-sized community health environments. Healthcare call centers frequently experience high call volumes, long hold times, and manual documentation burdens that can delay urgent triage decisions. This work presents the design and implementation of ClearPath Health, an AI-assisted voice intake and triage system developed in collaboration with Outer Cape Health Services (OCHS). The system replaces traditional integrated voice response (IVR) workflows with a structured conversational interface powered by Large Language Model (LLM) intent classification. This interface categorizes unscripted patient speech into emergency, same-day urgent, or non-urgent cases. Based on this autonomous classification, the system escalates high-risk cases and supports controlled appointment routing for non-emergent concerns via API integration. The objective is not to replace call center staff, but to reduce the cognitive load of human intake representatives and triage nurses by automating routine data collection and prioritizing high-acuity calls to improve response efficiency.

Keywords—AI in Healthcare, Voice Assistants, Medical Triage, Natural Language Processing, Call Center Automation

I. Introduction

Healthcare call centers serve as the primary access point for many patients seeking medical care, particularly in community health systems where phone-based communication remains dominant [6]. For example, in certain community health settings, including rural or underserved areas, phone-based communication remains an important access channel, particularly for patient populations that may prefer direct interaction or have limited engagement with digital tools. High call volumes during peak hours often lead to extended hold times, delayed triage decisions, and increased workload for call center representatives. Traditional Interactive Voice Response (IVR) systems which rely on rigid "press 1 for pharmacy" menus offer basic routing capabilities but lack the ability to capture structured clinical information or assist with dynamic triage prioritization [7]. Based on our research at Outer Cape Health Services (OCHS), we identified several operational challenges. These include long hold times, delayed identification of urgent cases during high call volumes, and frequent back-and-forth missed calls between nurses and patients, often referred to as “phone tag”. Additionally, staff experience a significant manual burden when entering patient concerns into electronic health systems; currently, a representative must manually transcribe a caller's symptoms into the patient record, a process that takes several minutes and is prone to human error. We illustrate the potential of ClearPath Health, an AI-assisted voice intake and triage system, to support and enhance these existing workflows. Unlike standard IVR systems, ClearPath Health utilizes a prompt-engineered LLM-based Natural Language Understanding (NLU) engine to perform real-time symptom extraction and intent mapping from patient speech. Rather than replacing human operators, the system is designed to support call center representatives by collecting structured patient information, categorizing urgency levels through a hybrid AI-assisted and rule-based approach, and facilitating appropriate escalation or scheduling actions. The goal is to improve efficiency and consistency in the intake process while maintaining safety through human oversight. This paper presents a voice-based, LLM-assisted triage prototype for community health settings that combines conversational intake, urgency classification, and action routing within a human-supervised workflow

II. Background and Related Work

The integration of Artificial Intelligence in clinical settings has transitioned from theoretical models to operational tools, particularly in the domain of medical triage and patient communication. Previous research has demonstrated that Machine Learning (ML) can significantly improve diagnostic accuracy and patient prioritization [3]. However, many enterprise-level AI solutions are designed for large hospital systems with extensive IT infrastructure, often making them inaccessible or overly complex for smaller community health centers like Outer Cape Health Services (OCHS).

Existing solutions generally fall into three categories:

1. **Traditional Interactive Voice Response (IVR)**: These systems utilize rigid, button-triggered menus (e.g., "Press 1 for Pharmacy"). While reliable, they provide no clinical intelligence and often increase patient frustration due to long navigation paths.
2. **General-Purpose LLM Chatbots**: Platforms like ChatGPT or Claude can engage in sophisticated medical dialogue but lack the native voice integration and the strict safety "guardrails" required for clinical triage. Without a structured

state-machine logic, these models are prone to "hallucinations" or inconsistent urgency classification [4].

3. **Enterprise Health AI Platforms**: Comprehensive platforms offer deep integration with Electronic Health Records (EHR) but require significant financial investment and technical overhead to implement.

ClearPath Health is positioned as a lightweight, hybrid triage assistant that bridges the gap between these categories. Unlike standard IVR, it uses Natural Language Understanding (NLU) to allow for unscripted conversation. Unlike general-purpose chatbots, it operates within a deterministic triage framework, specifically the three-tier OCHS protocol ensuring that the AI's output remains clinically grounded and safe. The system design of ClearPath Health is centered on an orchestration layer. By utilizing a high-performance voice-streaming platform (Vapi) integrated with a Large Language Model (LLM), the system enables real-time voice interaction while simultaneously performing complex data extraction and intent mapping [1]. This enables a "human-in-the-loop" model where the AI pre-processes the call, identifies the urgency level, and structures the data before it reaches a human nurse. This design follows a safety-first approach, ensuring that clinical decision-making remains under human supervision while improving efficiency in resource-constrained environments. In addition to these systems, prior research has explored the use of conversational agents for symptom assessment and patient intake in healthcare settings [7], [8]. These approaches often leverage natural language processing to guide patients through structured decision trees or symptom checkers. However, concerns remain regarding safety, transparency, and the potential for incorrect or overconfident medical recommendations when large language models are used without constraints. Recent work has emphasized the importance of human-in-the-loop designs and protocol-based safeguards when deploying AI in clinical workflows. Furthermore, multilingual support has become an important consideration in healthcare communication systems. Studies in multilingual clinical natural language processing have highlighted challenges in accurately interpreting patient-reported symptoms across languages and dialects [9]. In this context, the present work differs by implementing a voice-first, protocol-constrained triage system designed for a community health setting, incorporating human supervision and structured escalation mechanisms.

III. Project Description and Implementation

The ClearPath Health system is designed as a modular, cloud-native platform designed to bridge the gap between patient voice input and clinical action. The system architecture, Fig. 1, and the logical workflow, Fig. 2, illustrate the transition from unstructured natural language to structured data classification.

A. System Architecture

The architecture is divided into three primary layers: the Interface Layer, the Intelligence Layer, and the Action Layer.

- **Interface Layer**: Utilizes a high-concurrency voice-streaming gateway (Vapi) to manage WebRTC or PSTN connections. This layer handles real-time Speech-to-Text (STT) transcription using a deep-learning-based acoustic model.
- **Intelligence Layer**: This layer serves as the core reasoning component of the system. It consists of a Large Language Model (LLM) configured with a medical-specific orchestration prompt. Unlike traditional IVR, this layer performs Entity Extraction (capturing name, DOB, symptoms) and Intent Classification (mapping symptoms to OCHS protocols).
- **Action Layer**: Based on the LLM's classification, the system executes specific API tool-calls (eg, Google Calendar API for scheduling or a SIP-transfer for emergency nurses).

B. Triage Logic and System Design

A common limitation of traditional voice assistants is their reliance on rigid keyword-based or “if-then” logic. In contrast, ClearPath Health employs a hybrid AI-assisted approach in which the Large Language Model (LLM) is responsible for semantic interpretation of patient speech, while final triage decisions are governed by predefined clinical protocols. Specifically, the LLM processes unstructured patient input and extracts structured information such as symptoms, duration, and severity [9]. For example, a statement such as “I feel like an elephant is sitting on my chest” is interpreted as a potential cardiac-related symptom. However, the LLM does not independently assign a clinical urgency level. Instead, all triage decisions are constrained by the OCHS three-tier clinical protocol, which maps extracted symptoms to Category 1 (Emergency), Category 2 (Same-Day Urgent), or Category 3 (non-urgent) cases. This design ensures that AI is used strictly for natural language understanding, while medical decision-making remains deterministic, transparent, and aligned with established clinical guidelines. Additionally, a dynamic priority override mechanism is implemented, where certain high-risk symptom patterns (e.g., slurred speech, severe chest pain) immediately trigger escalation regardless of conversation state. This safety-first design prevents delays in emergency handling and ensures clinical reliability.

C. Technical Implementation

The prototype was implemented using a combination of Python-based microservices and cloud APIs.

1. **Natural Language Understanding (NLU)**: We utilized a Large Language Model (GPT-4.1 mini via Azure OpenAI) as the primary reasoning engine. The model was given a system prompt containing the OCHS Three-Tier Triage Framework.
2. **Backend Integration**: A Streamlit-based dashboard was developed for clinical oversight, allowing nurses to view real-time call logs, categorized symptoms, and recording transcripts. This provides the “Human-in-the-Loop” validation required for medical AI.

3. **API Orchestration**: We developed custom JSON schemas for tool-calling. For example, the same day appointment tool identifies the current date (using system variables) and queries available 30-minute slots via the Google Calendar API, ensuring appointments are never booked in the past.

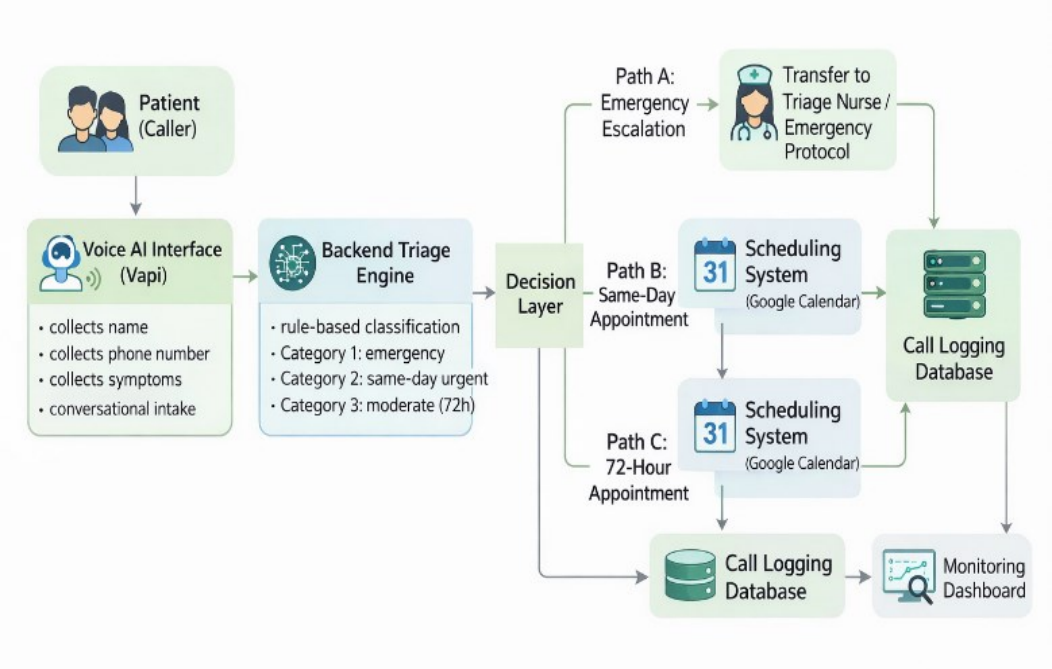

Fig. 1. System architecture of the AI-assisted medical triage system.

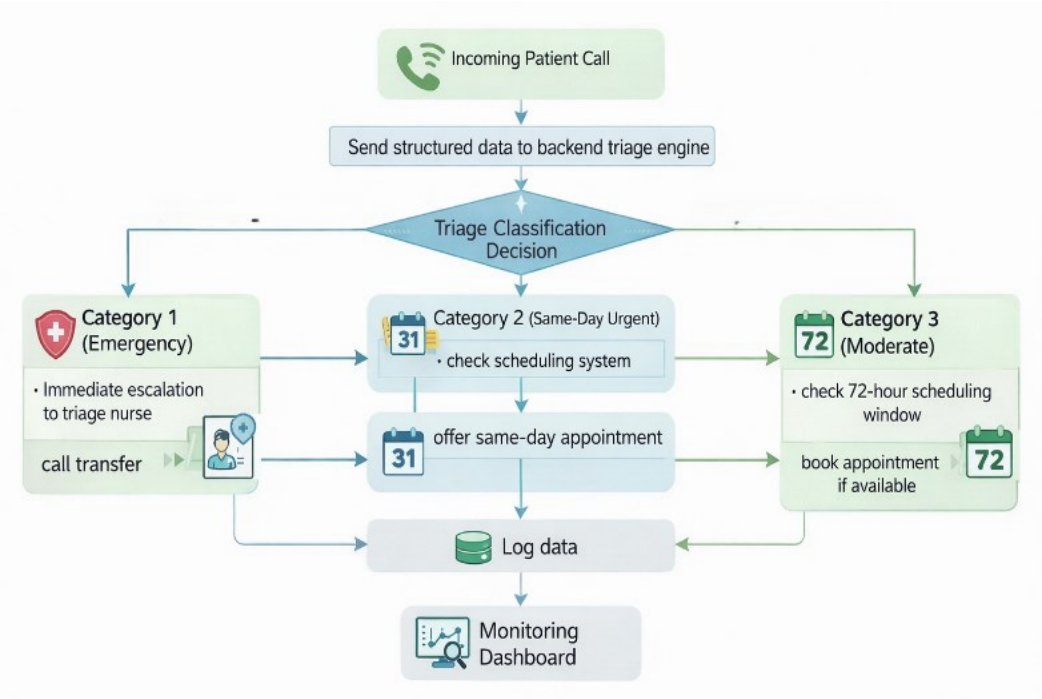

Fig. 2. Patient call workflow and triage decision process.

IV. Preliminary Results and Evaluation

To evaluate the performance and usability of the proposed system, we conducted a controlled pilot study involving 12 participants with diverse linguistic backgrounds. The participant group consisted of 5 native English speakers, 4 non-native English speakers with varying levels of fluency, and 3 native Spanish speakers. Participants were recruited from the student population and included individuals from the authors' extended network (e.g., peers, friends, and family members) to ensure diversity in language use and communication styles. Each participant was asked to

complete one or more simulated call interactions using predefined clinical scenarios corresponding to the three OCHS triage categories. These scenarios included both emergency and non-emergency cases (e.g., chest pain, abdominal pain, cough, rash) and were designed to evaluate the system's ability to interpret natural language descriptions of symptoms. Participants were instructed to describe symptoms either by following a short script or by paraphrasing the scenario in their own words. This approach allowed evaluation under both controlled and semi-natural conversational conditions.

A. Operational Evaluation Framework

Before analyzing the data, we established three primary criteria to justify the system's performance. These criteria were chosen to measure the system's ability to act as a reliable "first responder" in a clinical call center:

1. Correctness of triage classification, determined by comparing the system's output to the expected category defined by the OCHS protocol.
2. Completeness of entity extraction, defined as successful capture of required patient information (name, callback number, and symptoms)
3. Successful execution of the appropriate system action (e.g., escalation or scheduling).

All classifications were verified against the predefined OCHS clinical triage guidelines to ensure consistency in evaluation.

B. System Performance Results

The results of the 12-user pilot study are summarized in Table I. The system demonstrated high reliability in detecting high-acuity cases, even when symptoms were described using colloquial or non-technical language.

Table I: Performance Metrics Across 12 Randomized Pilot Scenarios

Language	Target Category	System Classified Category	Data Extraction Complete?	Outcome
English	Category 1 (Chest Pain)	Category 1	Yes	Success (Auto-Transfer)
English	Category 3 (Back Pain)	Category 3	Yes	Success (Scheduled)
Spanish	Category 2 (Fever/UTI)	Category 2	Yes	Success (Same-Day)
English	Category 1 (Stroke)	Category 1	Yes	Success (Auto-Transfer)
Spanish	Category 1 (Breathing)	Category 1	Yes	Success (Auto-Transfer)
English	Category 2 (Abdominal)	Category 2	Yes	Success (Same-Day)
English	Category 3 (Rash)	Category 3	Yes	Success (Scheduled)
English	Category 2 (Blood/Urine)	Category 2	Yes	Success (Same-Day)
English	Category 1 (Trauma)	Category 1	Yes	Success (Auto-Transfer)
English	Category 3 (Cough)	Category 3	Yes	Success (Scheduled)
English	Category 2 (Asthma)	Category 1 (Override)	Yes	Safety-Pass (Higher Triage)
Spanish	Category 2 (Flank Pain)	Category 2	Yes	Success (Same-Day)

C. Discussion of Results

In this 12-scenario pilot study, the system correctly identified all simulated Category 1 emergency cases. Notably, in Scenario P11, the AI categorized a non-urgent asthma exacerbation as Category 1 rather than Category 2. While this is technically an over-classification, it aligns with our "Safety-First" design principle, ensuring that any ambiguity in breathing difficulty is treated with maximum urgency. The inclusion of Spanish-speaking participants (P03, P05, P12) suggest that the system can support cross-lingual intake in controlled testing conditions without requiring separate codebases.

V. DISCUSSION AND LIMITATION

The preliminary results suggest that the proposed system can support structured voice-based triage in controlled testing conditions. The system demonstrated the ability to classify symptom descriptions and route cases according to the predefined OCHS triage protocol, while supporting both English and Spanish interactions during the pilot study. These findings indicate that conversational intake combined with protocol-based decision logic may offer potential advantages over traditional IVR workflows, particularly in terms of flexibility and completeness of data collection. However, several limitations should be considered. First, the evaluation was conducted as a small-scale pilot study with 12 participants using simulated scenarios rather than real patient calls. As a result, the findings may not fully reflect real-world variability in communication patterns, including background noise, interruptions, or incomplete symptom descriptions. Second, similar challenges are also present in human-led triage, where communication errors, incomplete descriptions, or misinterpretations can occur during patient interactions. While Spanish-language cases were supported during testing, broader language coverage and robustness remain areas for further study. Third, the system does not include a direct comparison against existing IVR workflows, limiting the ability to quantify relative performance improvements. Additionally, the safety-first design of the triage protocol may lead to over-classification of higher urgency categories. While this can increase the number of escalations, it reflects a deliberate "do no harm" approach, where missing a critical condition is considered

more severe than triggering an unnecessary escalation. Finally, the system currently operates as a prototype and does not fully integrate with enterprise Electronic Health Record (EHR) systems, which presents challenges for real-world deployment. The system also relies on prompt-based configuration rather than model fine-tuning, which may impact consistency under more complex conversational scenarios. Additionally, considerations around data privacy and secure handling of patient information remain critical for real-world deployment, particularly when integrating LLM-based systems into clinical environments.

VI. Conclusion

This paper presented ClearPath Health, a voice-based, LLM-assisted triage system designed to support patient intake in community healthcare settings. The system integrates conversational data collection with rule-based triage classification aligned with OCHS clinical guidelines, enabling structured intake while maintaining human oversight. Results from a controlled pilot study demonstrate that the system can successfully interpret patient-reported symptoms and route cases according to predefined triage categories in a conversational setting. While the findings are limited to a small-scale evaluation, they suggest that AI-assisted voice intake systems have the potential to enhance call center workflows by improving intake efficiency and reducing manual data entry. Future work will focus on expanding evaluation to larger and more diverse populations, improving robustness in multilingual and noisy environments, and integrating the system with real-world clinical infrastructure to assess operational impact in practice.

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